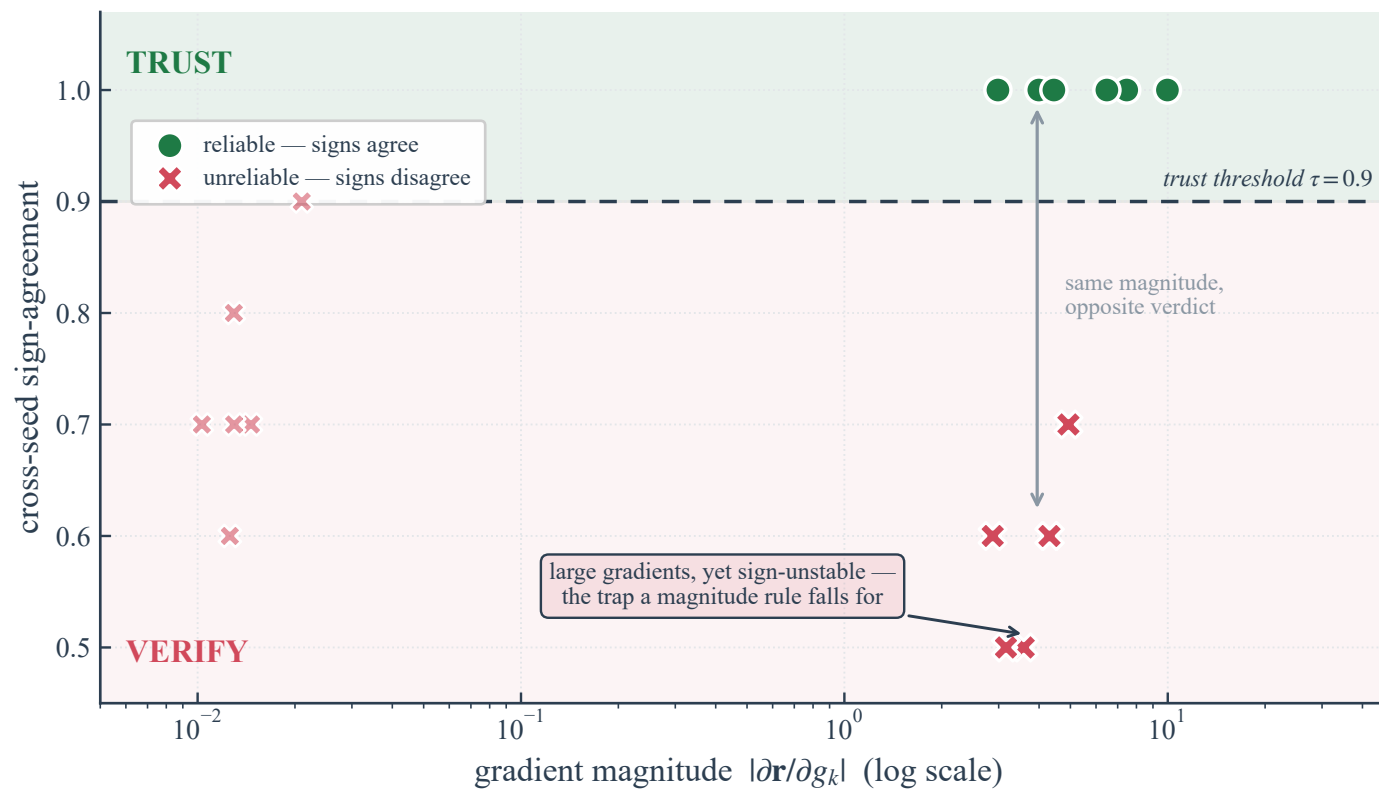


Direction, not magnitude

which design gradients of a differentiable surrogate to trust — set by cross-seed sign agreement, not gradient size



● **signs agree → TRUST**
use the free autodiff gradient (0 solves)

● **signs disagree → VERIFY**
spend one full-wave solve

solver-free reliability gate

AUC 0.91

predicts gradient sign-correctness
on an external transfer-matrix benchmark